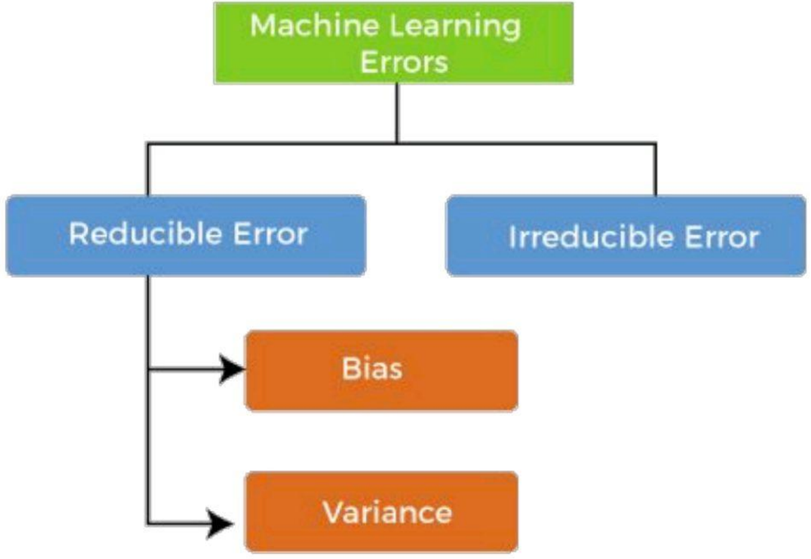
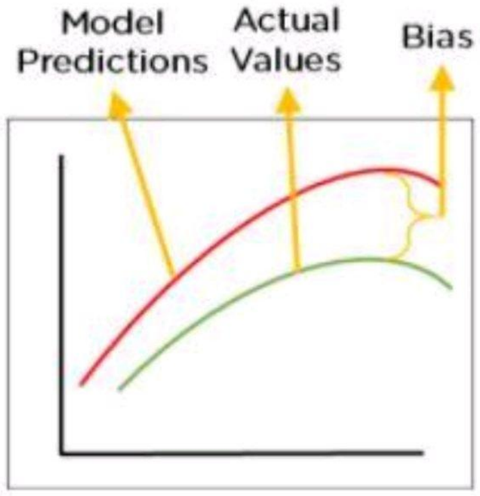
Error in Machine Learning



Error in Machine Learning

Bias

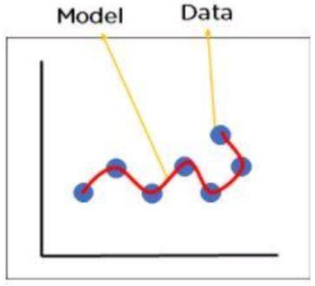
- Sai lệch giữa prediction và actual label
- Cho ta biết khả năng dự đoán chính xác của mô hình
- Bias càng bé càng tốt
- High bias means:
  - Overly-simplified model
  - Under-fitting
  - High error on both training and test sets

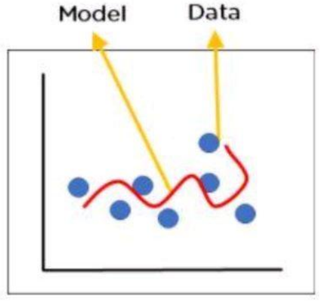


Error in Machine Learning

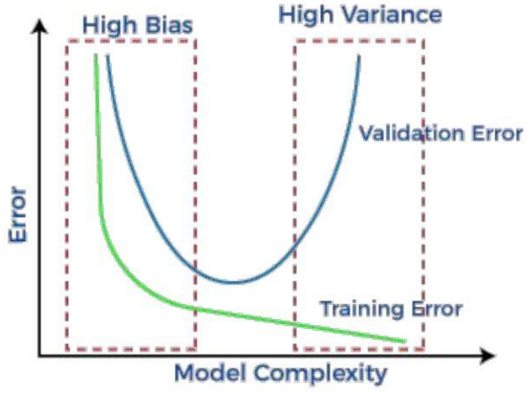
Variance

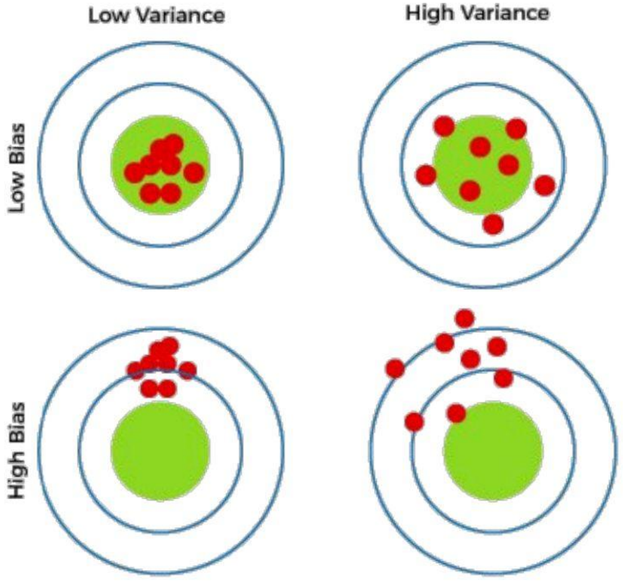
- Mức độ thay đổi của độ chính xác của prediction nếu input thay đổi
- Cho ta biết khả năng tổng quát hóa của mô hình
- Variance càng bé càng tốt
- High variance means:
  - Overly-complex model
  - Over-fitting
  - Low error on training set but high error on test set

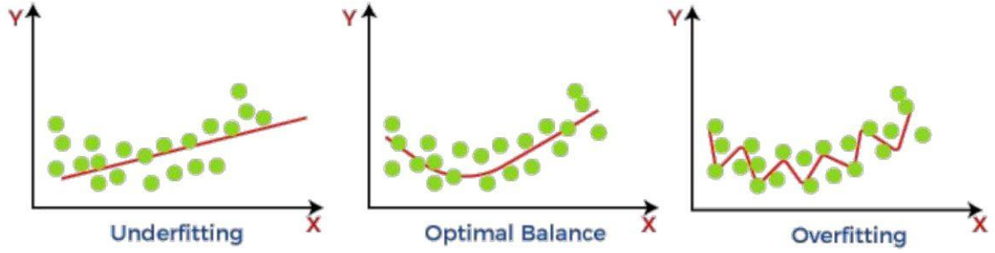


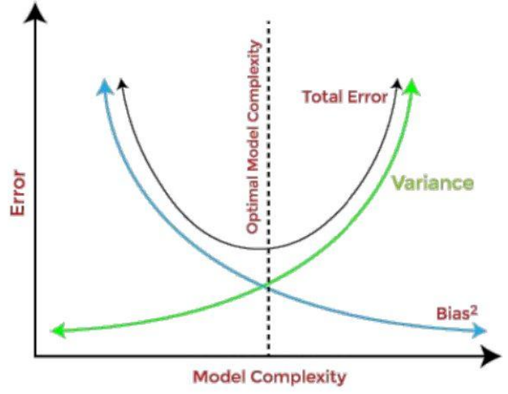


Bias-Variance tradeoff






Y
X
Underfitting
Y
X
Optimal Balance
Y
X
Overfitting


Error
Optimal Model Complexity
Total Error
Variance
Bias²
Model Complexity

Choose Machine Learning model based on data

Low bias + high variance

- Decision tree
- Random Forest
- K-nearest neighbours
- Kernel SVM

-> Phù hợp khi có nhiều data và ít features

High bias + low variance

- Linear regression
- Logistic regression
- Linear SVM

-> Phù hợp khi có ít data và nhiều features

Extra

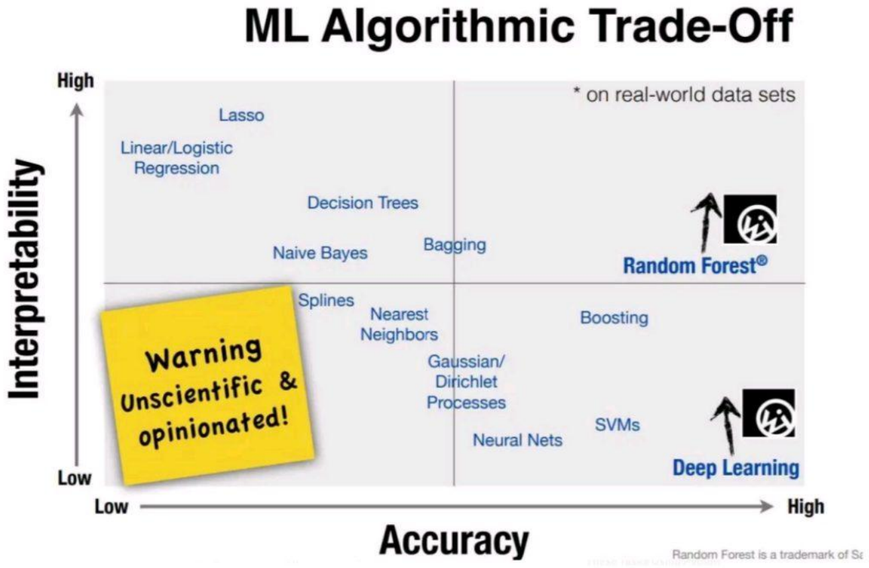
Nếu có nhiều data và nhiều features

- Giảm chiều dữ liệu (e.g. PCA)
- Sử dụng Neural network

Choose Machine Learning model based on business domain

- **Which metric is important?**
  - Accuracy, precision, recall, F1, ...
- **What is the priority?**
  - Speed: self-driving car app need to be realtime -> fast
  - Memory: Model deployed in embedded device needs to be small
  - Accuracy: In medical field, accuracy is the most important factor
- **Interpretability ?**
  - Do you need to explain result?
  - Do you need to find out which features are important?

Machine Learning algorithm's interpretability



Tips for choosing Machine Learning model

- Start with a simple model
  - Choose the simplest model first
  - If it is good enough, you even do not need to try another model
- Try different models and shortlist the best ones?
- Do Hyperparameter Tuning for each models?
  - GridSearchCV if the number of combinations is small
  - RandomizedSearchCV if the number of combination is large
- Compare amongst the best models with best hyperparameters to pick up the best one

Underfitting and Overfitting

